HENRY SHIRES

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OBJECTIVE

Computer Engineer with professional software and AI experience, seeking a hands-on role in embedded systems and hardware/software integration.

EDUCATION

B.S. Computer Engineering, Entrepreneurship Minor | Graduated May 2025 | Iowa State University | 3.87 GPA
University Honors Program, Dean's List Fall 2021 – Spring 2025

EXPERIENCE

AI Engineer | Zirous, Inc. (VAEZR Studio) | West Des Moines, IA | May 2025 – Present

- Delivered AI enablement and transformation solutions for clients, from proof of concept through implementation
- Built generative AI and retrieval-augmented generation (RAG) solutions and integrated tools and data sources using Model Context Protocol (MCP)
- Developed full-stack applications spanning frontend UI, backend services, and data layers to bring AI features into client products
- Served as primary developer on a complex AI-assisted proof of concept, translating client requirements into a successful demonstration
- Delivered an AI-powered experience for a large public university, nominated for a Technology Association of Iowa (TAI) Prometheus Award in CX/UX

Embedded Software Engineer Intern | Vermeer Corporation | Pella, IA | May – August 2024

- Collaborated with the Intelligent Systems team to build a mobile app connected to machine telematics data, shipped to the App Store for customers with Vermeer hay balers
- Designed and ran automated integration tests across application backend infrastructure and frontend UI to validate system behavior end to end
- Gained hands-on exposure to autonomy systems, including ROS 2
- Presented integration testing and app work at the company intern fair

Integration Team | CySat NASA CubeSat, Make to Innovate | Iowa State University | August 2022 – March 2024

- Collaborated with interdisciplinary engineers to build a satellite that launched in partnership with NASA in 2024 on a 3-month scientific mission
- Developed embedded software and radio communications on space-rated hardware to control satellite functions and mission tasks, integrating software with physical systems

Experience Engineer Intern | Zirous, Inc. (VAEZR Studio) | West Des Moines, IA | May – August 2023

- Solved engineering challenges involving head-mounted displays (HMDs) and 3D game development
- Designed visual experiences with the Unity 3D game engine and wrote game object functionality in C#
- Developed and presented to a client a re-creation of their manufacturing plant in a VR tour

Co-founder, Chief Executive Officer | Casmium, LLC | West Des Moines, IA | February 2020 – Present

- **Youth Baseball Analytics Software-as-a-Service:** Combined entrepreneurial visions, software engineering, team management, and other fundamental operations to pursue an innovative business venture
- Managed software development sprints and issues to ensure a high-quality and reliable MVP
- Scaled to 25+ teams scoring 250+ games and 3 million+ data points in a single season
- Won over $25,000 in pitch competition prizes, reinvested to fund product development and growth
- Partnered with Bound, an Ames-based company, as a software API integrator to upload game statistics to their platform

SKILLS

Embedded & Hardware: C/C++, ARM, GPIO, VHDL, Verilog, MIPS Assembly, Intel Quartus Prime, QuestaSim (ModelSim)
Software: JavaScript (Node.js, React.js, React Native), Python, Java, C#, SQL, NoSQL, HTML/CSS, Git, Appium
AI: Generative AI, RAG, MCP, Microsoft Foundry, GitHub Copilot, LangChain
Other Technologies: Azure, AWS, Google Cloud, Figma, GitHub, JetBrains, Microsoft 365, Atlassian, Docker

PROJECTS & ACTIVITIES

Labs and Projects | Iowa State University | 2022 – 2025

- **Senior Design:** Team capstone project in which microcontrollers interface with a web UI via TCP and HTTP
- **Embedded Systems Lab:** Programmed a TI Tiva ARM Cortex microcontroller with sensors in C and ARM Assembly to prototype a small-scale self-driving bus on an iRobot Roomba platform
- **Computer Architecture Lab:** Designed three MIPS processors (single-cycle and pipelined) in VHDL, simulated them in QuestaSim, and compared synthesized clock speeds
- **Machine Learning Hardware Lab:** Designed a CNN accelerator in VHDL, then benchmarked TensorFlow, PyTorch, and scikit-learn running the same network on it to determine which framework performed best (Python, C++)
- **Digital Logic Lab:** Designed an FPGA-based register-sorting circuit in Verilog on an Altera DE2-115 board using Intel Quartus Prime

CYstarters Business Accelerator & Pitch Competitions | ISU Pappajohn Center for Entrepreneurship | Ames, IA | May 2022 – March 2023

- Conducted customer discovery interviews to understand customer problems, evaluate current solutions, and translate customer needs into clear product requirements
- Determined value proposition and go-to-market strategy based on coaches' testing feedback
- Pitched Casmium, LLC to the Okoboji Entrepreneurial Institute (2022) and ISU College-by-College Pitch-off (2023)

TreeHacks 10 Hackathon Participant | Stanford University | February 2024

- Built the frontend, using Next.js, Convex, and Ant Design, for Spark, a social platform that connects volunteers with organizations' campaigns, on a four-person team in 36 hours
- Shaped features through an interview with a nonprofit executive director, and presented a working end-to-end MVP to judges
- Won 3 sponsor awards: Convex (Best Use of Convex Features), Stanford Ecopreneurship (Best Prototyping Process), and Cotopaxi (Most Innovative Sustainability Hack)

Iowa State University Cyclone Football 'Varsity' Marching Band, Basketball Pep Band | August 2021 – May 2025

- Varsity Drumline Section Leader (Bass Drum): Took leadership responsibilities to teach drumline fundamentals, manage logistics, and maintain a safe and fun environment for band members to thrive in